Ownership Transfers and Eviction Filing Rates

Philly Evictions Project

2026-03-28

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1 Overview

This note reports the streamlined annual transfer-evictions specification. The main outcome is `filing_rate = num_filings / total_units` in a full building-year panel from 2006 to 2019, with single-transfer treated buildings compared against buildings never transferred in the RTT data. All annual event studies are estimated on `t \in [r event_window_min, r event_window_max]`, with omitted event time `t = r event_ref_period`.

For filing outcomes, the main specification is a weighted stacked event study:

$$y_{iat} = \sum_{k \in \{-3,-1,0,1,2,3\}} \beta_k \mathbf{1}\{\text{rel_time}_{ia} = k\} + \alpha_{ia} + \lambda_{at} + \varepsilon_{iat},$$

estimated on a cohort-stacked panel with weights `q_weight × total_units`, where `q_weight` is the Wing et al. corrective control weight and `\alpha_{ia}` and `\lambda_{at}` are stack-specific parcel and year fixed effects. Non-filing outcomes continue to use the annual `PID + year` event-study design on the same `-3:3` window.

The annual surface has been pruned to four blocks:

- filing-rate event studies
- occupancy, resident-composition, complaint, and permit outcomes
- rent outcomes using the unified and safe series

- rental-evidence event studies for `1-unit` and `2+ unit` buildings

Portfolio filing types are computed using the helper logic from the production script, with portfolio size and filing rates defined using all holdings, including `1-unit` buildings. The filing and neighborhood-outcome regressions then restrict the estimation sample as needed.

2 Main Filing Results

The main filing figures now use the weighted stacked estimator directly. The filing sample grid separates `1-unit` properties from the multifamily stock. The main justification for the `2+ unit` analysis sample is that the `2+` and `5+` estimates are close, while `1-unit` properties are shown separately as the single-family contrast and the `10+` cut is treated as a stricter appendix validation rather than the main specification.

At `t = +1`, the filing-rate coefficient is 0.0096 for `2+ units`, 0.0059 for `5+ units`, and 0.0030 for `10+ units`.

Table 1: Sample sizes for the streamlined annual filing-rate event studies

Sample	Rows	Treated PIDs	Control PIDs
All buildings	6,960,484	32,006	123,376
1-unit	5,148,966	24,676	90,413
2+ units	1,811,495	7,330	32,976
2+ units, rental evidence	985,907	6,148	28,792
5+ units	258,330	662	5,089
5+ units, rental evidence	113,778	548	3,874
10+ units	121,631	290	2,551
10+ units, rental evidence	57,728	236	1,923

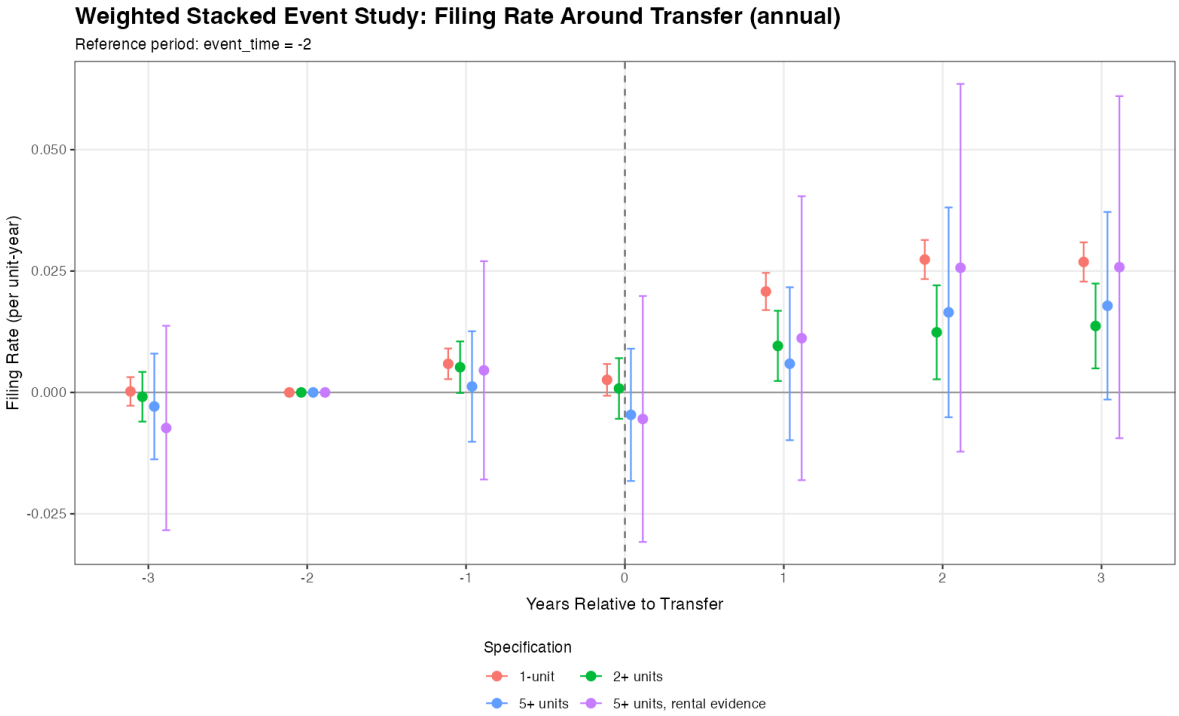

Figure 1: Weighted stacked annual filing-rate event study by sample restriction. The main figure overlays 1-unit, 2+, 5+, and 5+ rental-evidence samples; 10+ remains a stricter appendix check in the tables.

					filing_rate			
Sample	All buildings (1)	1-unit (2)	2+ units (3)	2+ units, rental evidence (4)	5+ units (5)	5+ units, rental evidence (6)	10+ units (7)	10+ units, rental evidence (8)
st_et_m3	-0.0005	0.0002	-0.0009	-0.0051	-0.0029	-0.0073	-0.0039	-0.0100
	(0.0015)	(0.0015)	(0.0026)	(0.0058)	(0.0056)	(0.0107)	(0.0068)	(0.0128)
st_et_m1	0.0056***	0.0059***	0.0052*	0.0125**	0.0012	0.0045	0.0017	0.0018
	(0.0016)	(0.0016)	(0.0027)	(0.0063)	(0.0058)	(0.0115)	(0.0071)	(0.0132)
st_et_0	0.0017	0.0026	0.0008	-0.0017	-0.0046	-0.0055	-0.0100	-0.0164
	(0.0018)	(0.0017)	(0.0032)	(0.0070)	(0.0070)	(0.0129)	(0.0083)	(0.0150)
st_et_1	0.0151***	0.0208***	0.0096***	0.0099	0.0059	0.0112	0.0030	0.0052
	(0.0021)	(0.0020)	(0.0037)	(0.0084)	(0.0080)	(0.0149)	(0.0097)	(0.0173)
st_et_2	0.0195***	0.0274***	0.0124**	0.0113	0.0165	0.0257	0.0104	0.0153
	(0.0027)	(0.0021)	(0.0049)	(0.0106)	(0.0110)	(0.0193)	(0.0128)	(0.0218)
st_et_3	0.0199***	0.0269***	0.0137***	0.0115	0.0179*	0.0258	0.0127	0.0169
	(0.0025)	(0.0021)	(0.0045)	(0.0099)	(0.0099)	(0.0180)	(0.0114)	(0.0202)
Observations	6,960,484	5,148,966	1,811,495	985,907	258,330	113,778	121,631	57,728
R^2	0.44371	0.32750	0.62041	0.64365	0.78199	0.77970	0.78786	0.77942
Within R^2	8×10^{-5}	0.00010	6.61×10^{-5}	5.08×10^{-5}	0.00020	0.00028	0.00017	0.00026
pid_cohort fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
yr_cohort fixed effects	✓	✓	✓	✓	✓	✓	✓	✓

Table 2: Selected filing-rate coefficients from the streamlined sample grid

Sample	t+1	t+2	t+3
1-unit	0.0208	0.0274	0.0269
10+ units	0.0030	0.0104	0.0127
10+ units, rental evidence	0.0052	0.0153	0.0169
2+ units	0.0096	0.0124	0.0137
2+ units, rental evidence	0.0099	0.0113	0.0115
5+ units	0.0059	0.0165	0.0178
5+ units, rental evidence	0.0112	0.0257	0.0258

3 Filing Heterogeneity by Acquirer Type

The main heterogeneity split uses four acquirer types:

- `High-filer portfolio`
- `Low-filer portfolio`
- `Small portfolio`
- `Single-purchase`

These categories are defined from the helper-function leave-one-out portfolio measures, while the regression itself is restricted to `2+ unit` buildings. In the current annual weighted stacked outputs, the `t = +2` coefficient is 0.0460 for high-filer portfolios and -0.0115 for low-filer portfolios.

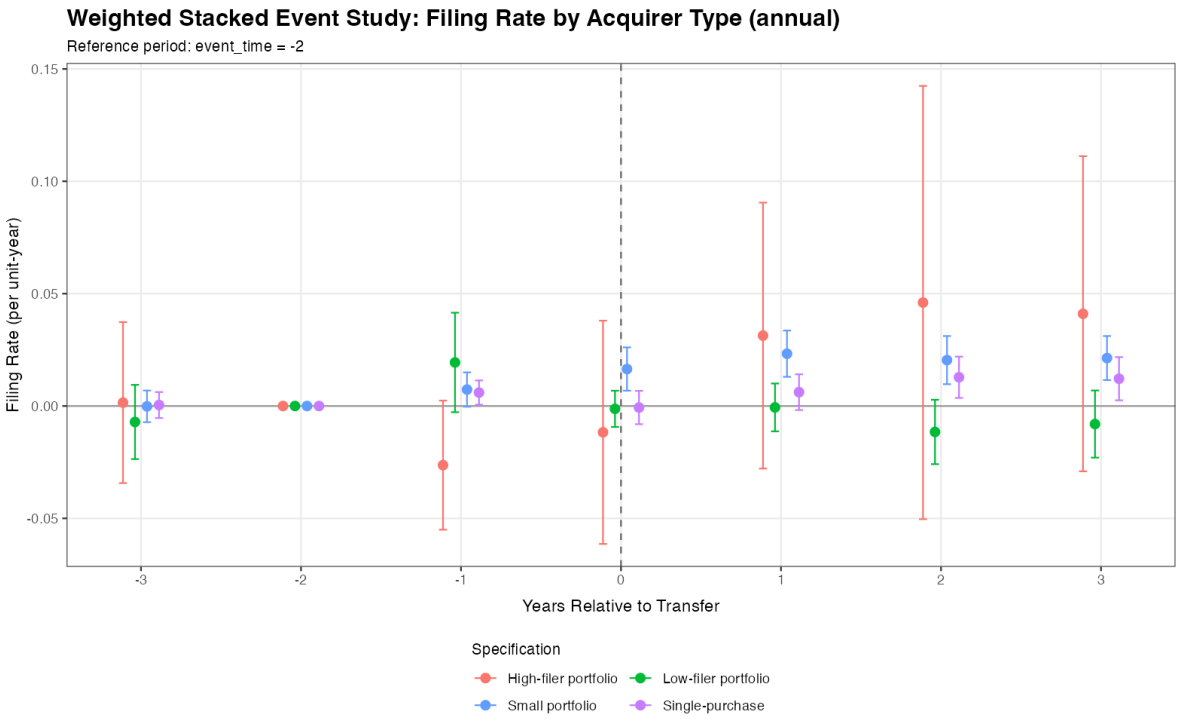

Figure 2: Annual filing-rate event study for 2+ unit buildings, split by acquirer filing type. Controls are never-transferred buildings in the same 2+ unit sample.

	filing_rate			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
st_et_m3	0.0015	-0.0071	-0.0002	0.0004
	(0.0183)	(0.0084)	(0.0036)	(0.0029)
st_et_m1	-0.0263*	0.0194*	0.0073*	0.0060**
	(0.0147)	(0.0113)	(0.0039)	(0.0028)
st_et_0	-0.0117	-0.0013	0.0165***	-0.0007
	(0.0254)	(0.0041)	(0.0049)	(0.0038)
st_et_1	0.0313	-0.0006	0.0233***	0.0062
	(0.0302)	(0.0054)	(0.0053)	(0.0041)
st_et_2	0.0461	-0.0115	0.0204***	0.0128***
	(0.0492)	(0.0073)	(0.0055)	(0.0047)
st_et_3	0.0410	-0.0080	0.0213***	0.0121**
	(0.0358)	(0.0076)	(0.0050)	(0.0049)
Observations	1,764,943	1,767,318	1,772,560	1,792,808
R^2	0.62451	0.62374	0.62118	0.62185
Within R^2	0.00010	2.63×10^{-5}	2.92×10^{-5}	3.61×10^{-5}
pid_cohort fixed effects	✓	✓	✓	✓
yr_cohort fixed effects	✓	✓	✓	✓

Table 3: Selected 2+ unit filing-rate coefficients by acquirer type

Acquirer type	t+1	t+2	t+3
High-filer portfolio	0.0313	0.0460	0.0410
Low-filer portfolio	-0.0006	-0.0115	-0.0080
Single-purchase	0.0062	0.0128	0.0122
Small portfolio	0.0233	0.0204	0.0213

4 Occupancy, Demographics, Complaints, and Permits

These outcomes are estimated on the `2+ unit` sample and split by the same acquirer-type bins as the filing regressions. The output surface here is intentionally narrower than the earlier report: the focus is occupancy, resident composition, complaints, and permits.

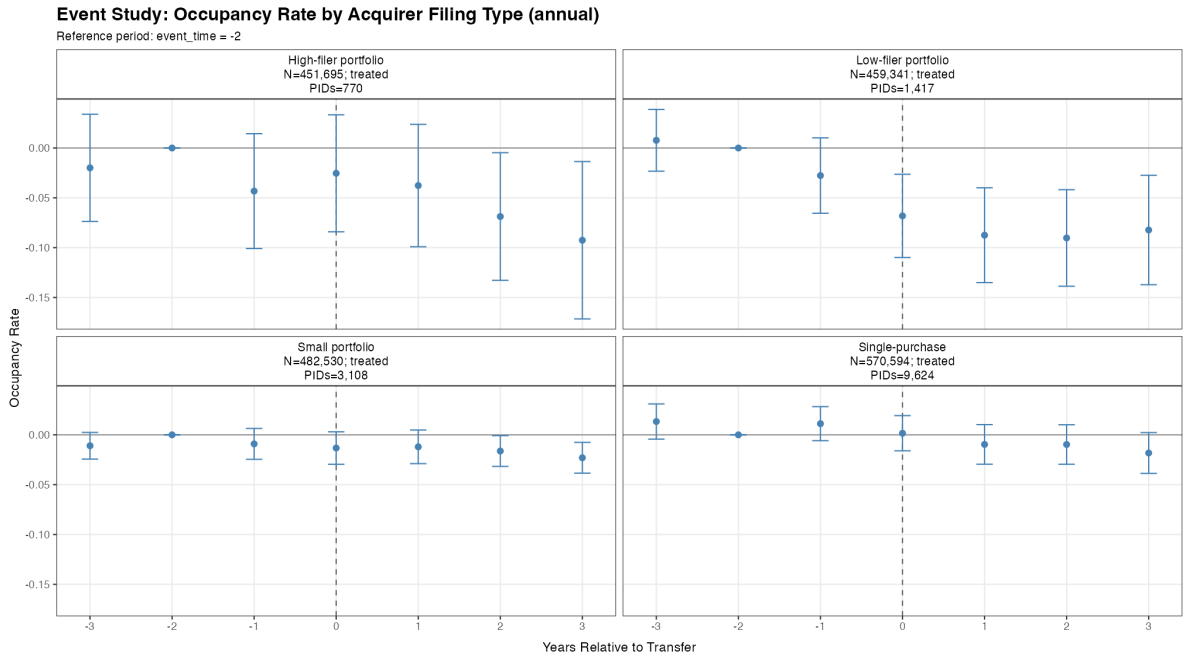

Figure 3: Occupancy-rate event study for 2+ unit buildings, overall and by acquirer filing type.

	occupancy_rate			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.0199	0.0077	-0.0110	0.0133
	(0.0274)	(0.0158)	(0.0068)	(0.0090)
et_m1	-0.0432	-0.0277	-0.0091	0.0112
	(0.0294)	(0.0193)	(0.0079)	(0.0087)
et_0	-0.0254	-0.0681***	-0.0133	0.0016
	(0.0299)	(0.0213)	(0.0083)	(0.0090)
et_1	-0.0377	-0.0875***	-0.0120	-0.0096
	(0.0313)	(0.0243)	(0.0086)	(0.0101)
et_2	-0.0687**	-0.0903***	-0.0163**	-0.0097
	(0.0327)	(0.0247)	(0.0079)	(0.0101)
et_3	-0.0925**	-0.0823***	-0.0230***	-0.0182*
	(0.0403)	(0.0280)	(0.0079)	(0.0104)
Observations	451,695	459,341	482,530	570,594
R^2	0.49360	0.50053	0.49623	0.50568
Within R^2	0.00037	0.00095	4.74×10^{-5}	0.00016
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

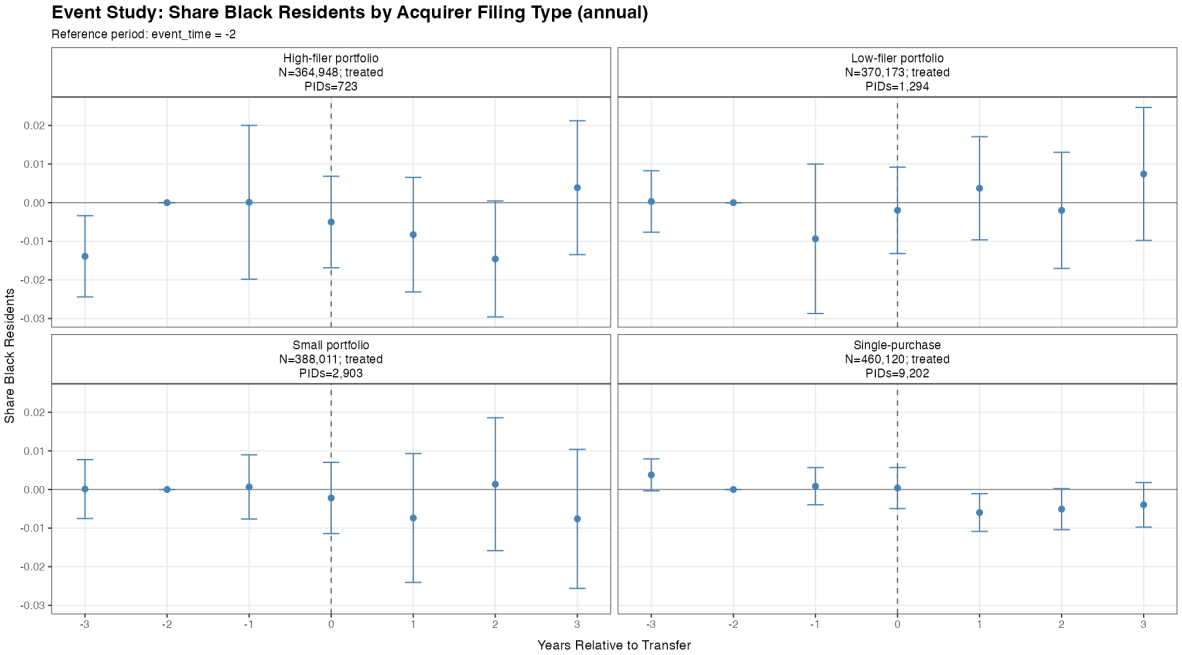

Figure 4: Share Black residents for 2+ unit buildings, overall and by acquirer filing type.

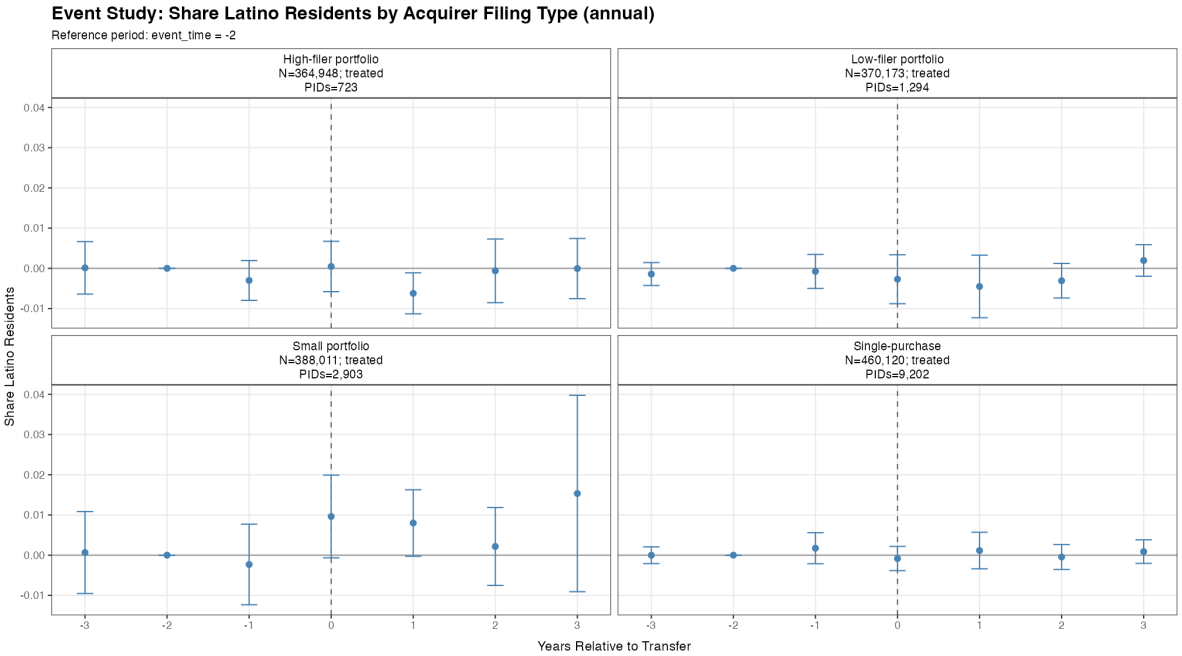

Figure 5: Share Latino residents for 2+ unit buildings, overall and by acquirer filing type.

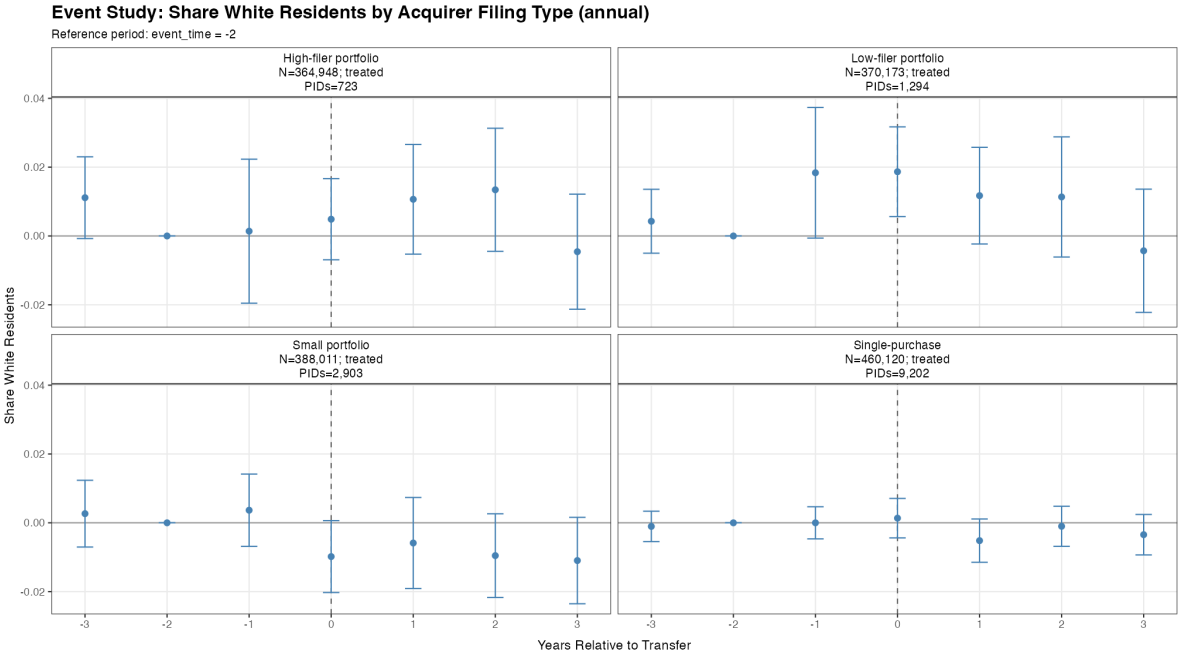

Figure 6: Share White residents for 2+ unit buildings, overall and by acquirer filing type.

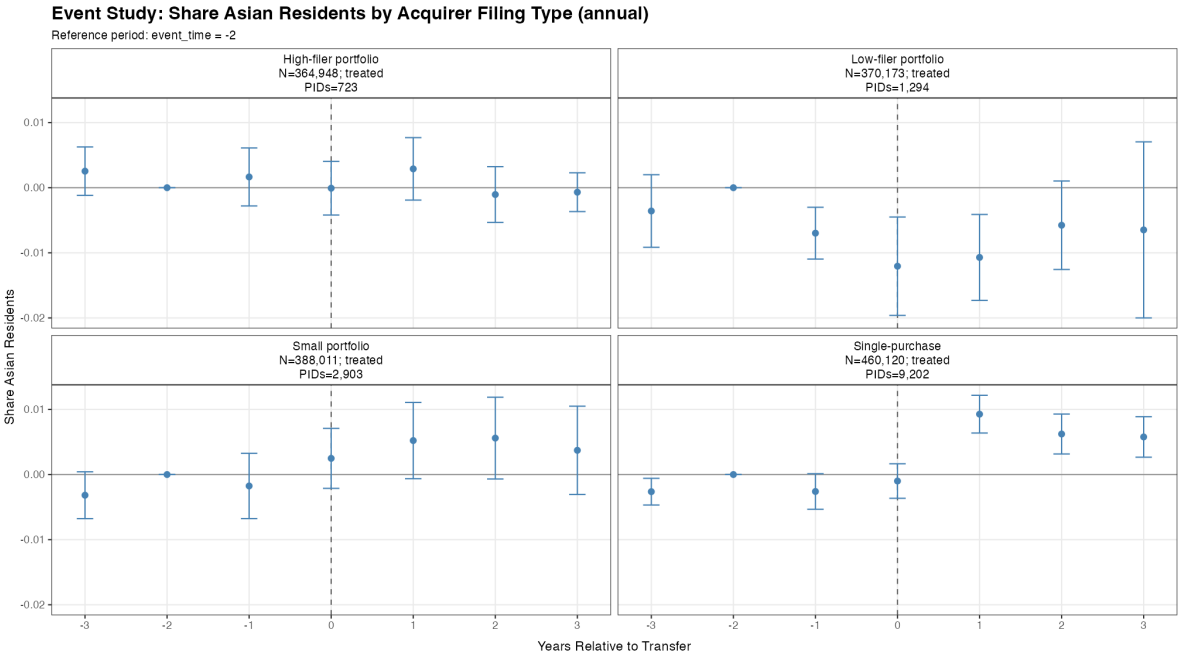

Figure 7: Share Asian residents for 2+ unit buildings, overall and by acquirer filing type.

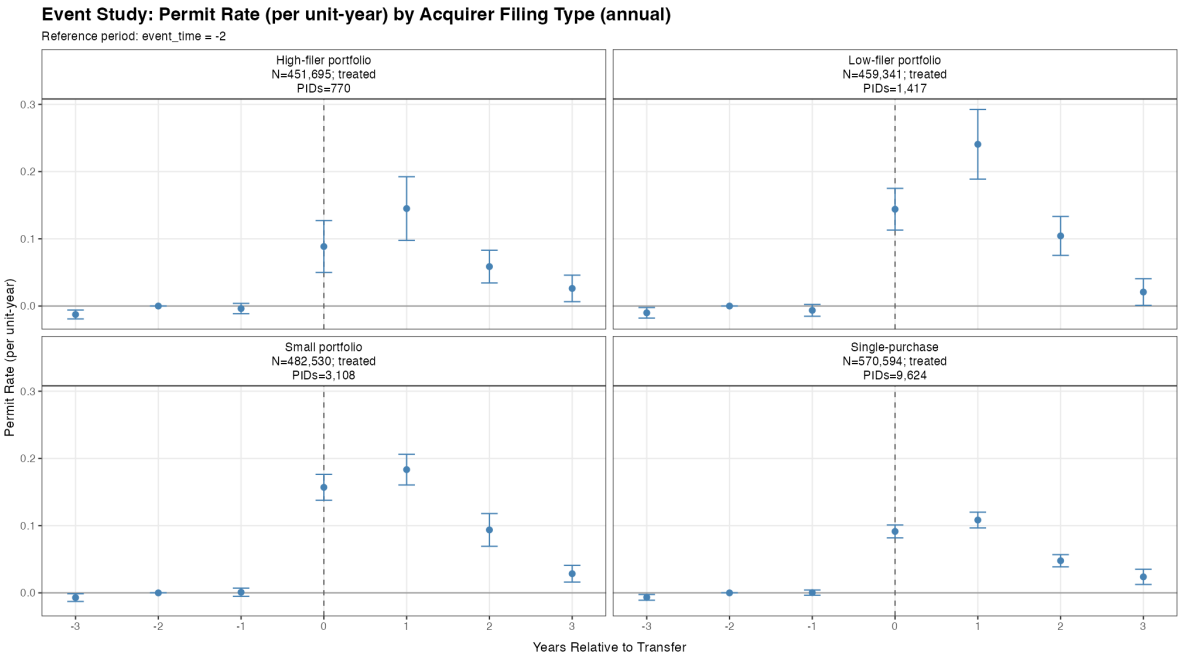

Figure 8: Permit-rate event study for 2+ unit buildings, overall and by acquirer filing type.

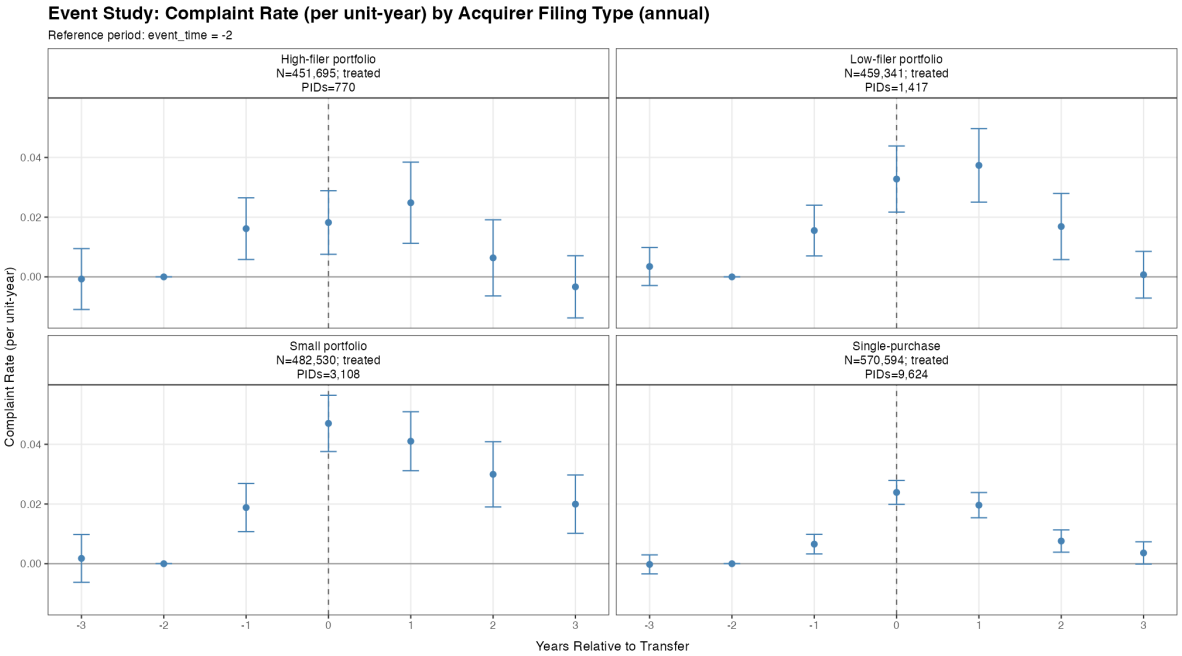

Figure 9: Complaint-rate event study for 2+ unit buildings, overall and by acquirer filing type.

5 Rent Outcomes

The rent analysis is run on all units, but treated buildings are required to have both pre- and post-acquisition support in the relevant rent measure. The report keeps both the unified rent series and the safe single-source series.

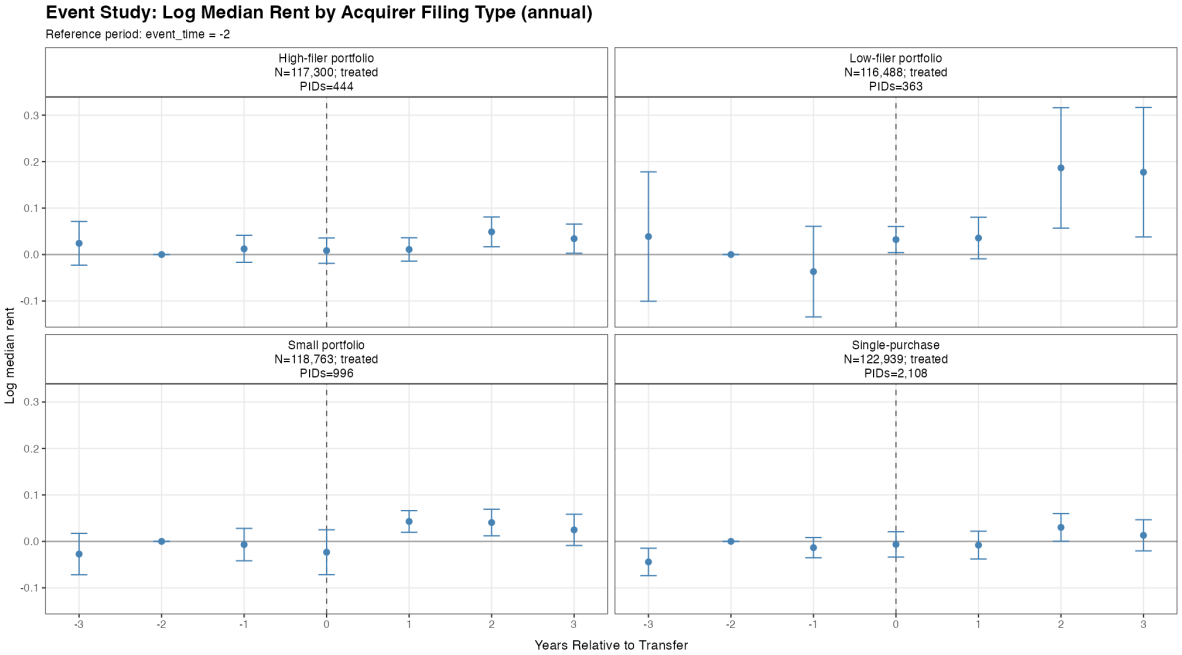

Figure 10: Log median rent event study, all units, overall and by acquirer filing type.

	log_med_rent			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	0.0242	0.0388	-0.0273	-0.0442***
	(0.0240)	(0.0710)	(0.0227)	(0.0151)
et_m1	0.0123	-0.0367	-0.0069	-0.0135
	(0.0148)	(0.0498)	(0.0178)	(0.0111)
et_0	0.0084	0.0323**	-0.0233	-0.0064
	(0.0139)	(0.0143)	(0.0246)	(0.0139)
et_1	0.0110	0.0356	0.0429***	-0.0080
	(0.0128)	(0.0228)	(0.0119)	(0.0152)
et_2	0.0488***	0.1865***	0.0406***	0.0301**
	(0.0163)	(0.0661)	(0.0146)	(0.0152)
et_3	0.0342**	0.1772**	0.0248	0.0131
	(0.0160)	(0.0711)	(0.0172)	(0.0171)
Observations	117,300	116,488	118,763	122,939
R^2	0.88831	0.88645	0.88769	0.88983
Within R^2	0.00041	0.00418	0.00034	0.00092
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

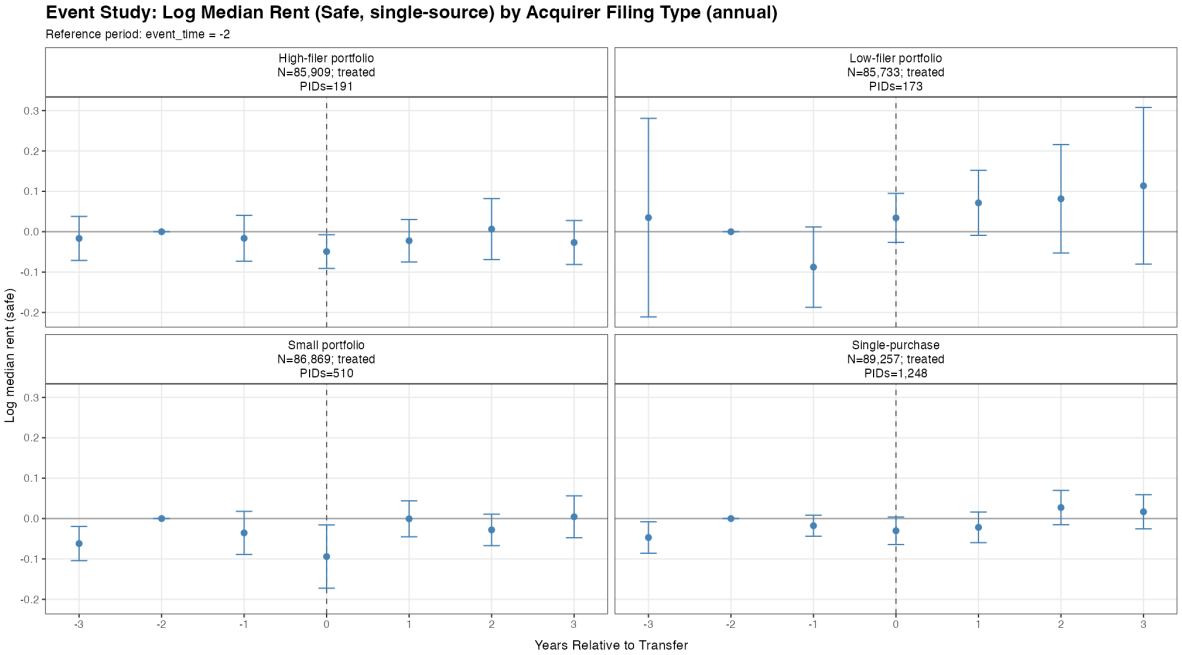

Figure 11: Log median rent (safe single-source series) event study, all units, overall and by acquirer filing type.

	log_med_rent_safe			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.0166	0.0349	-0.0619***	-0.0469**
	(0.0278)	(0.1255)	(0.0216)	(0.0199)
et_m1	-0.0163	-0.0877*	-0.0355	-0.0177
	(0.0290)	(0.0508)	(0.0272)	(0.0133)
et_0	-0.0492**	0.0343	-0.0940**	-0.0302*
	(0.0213)	(0.0310)	(0.0400)	(0.0174)
et_1	-0.0223	0.0715*	-0.0006	-0.0217
	(0.0268)	(0.0411)	(0.0227)	(0.0193)
et_2	0.0065	0.0816	-0.0281	0.0273
	(0.0385)	(0.0685)	(0.0199)	(0.0217)
et_3	-0.0267	0.1138	0.0044	0.0169
	(0.0278)	(0.0990)	(0.0265)	(0.0215)
Observations	85,909	85,733	86,869	89,257
R^2	0.91663	0.91568	0.91662	0.91800
Within R^2	0.00035	0.00190	0.00074	0.00160
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

6 Rental Evidence

These regressions show the probability of observed rental evidence separately for `1-unit` and `2+ unit` buildings. This is the direct check on the concern that transfers mechanically enter properties into the observed rental market.

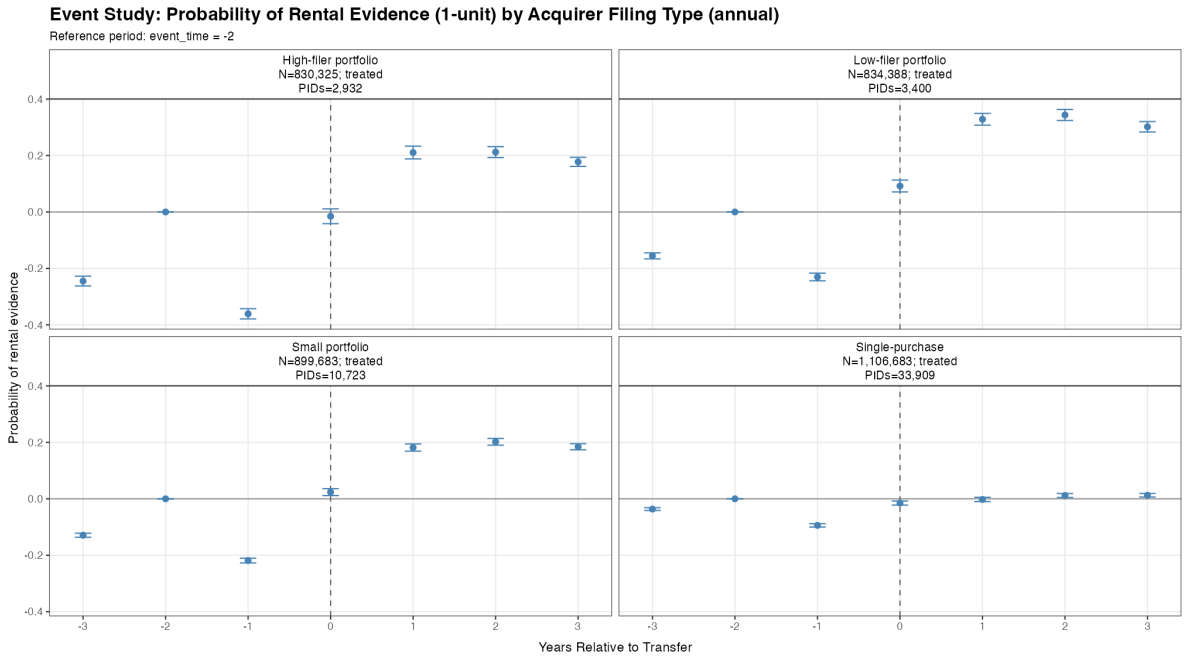

Figure 12: Probability of rental evidence for 1-unit buildings, overall and by acquirer filing type.

	has_rental_evidence			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.2448***	-0.1555***	-0.1291***	-0.0364***
	(0.0089)	(0.0054)	(0.0037)	(0.0025)
et_m1	-0.3609***	-0.2304***	-0.2187***	-0.0941***
	(0.0092)	(0.0069)	(0.0043)	(0.0030)
et_0	-0.0152	0.0920***	0.0237***	-0.0149***
	(0.0134)	(0.0107)	(0.0063)	(0.0037)
et_1	0.2107***	0.3284***	0.1817***	-0.0022
	(0.0115)	(0.0106)	(0.0065)	(0.0039)
et_2	0.2122***	0.3437***	0.2020***	0.0118***
	(0.0099)	(0.0100)	(0.0061)	(0.0036)
et_3	0.1775***	0.3018***	0.1845***	0.0127***
	(0.0082)	(0.0094)	(0.0055)	(0.0032)
Observations	830,325	834,388	899,683	1,106,683
R^2	0.65132	0.64838	0.64408	0.62180
Within R^2	0.00662	0.01061	0.01329	0.00169
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

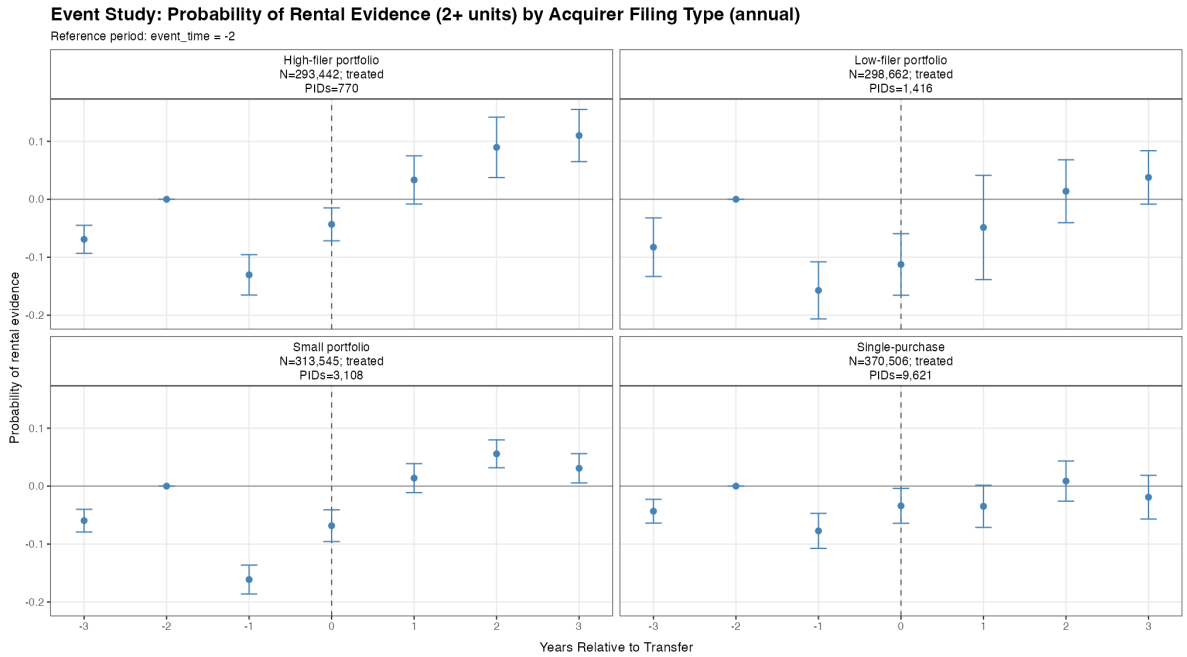

Figure 13: Probability of rental evidence for 2+ unit buildings, overall and by acquirer filing type.

	has_rental_evidence			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.0691***	-0.0827***	-0.0596***	-0.0433***
	(0.0123)	(0.0258)	(0.0100)	(0.0105)
et_m1	-0.1304***	-0.1572***	-0.1612***	-0.0772***
	(0.0178)	(0.0251)	(0.0127)	(0.0154)
et_0	-0.0433***	-0.1125***	-0.0683***	-0.0340**
	(0.0145)	(0.0271)	(0.0140)	(0.0154)
et_1	0.0334	-0.0486	0.0138	-0.0349*
	(0.0212)	(0.0459)	(0.0128)	(0.0186)
et_2	0.0896***	0.0138	0.0557***	0.0087
	(0.0266)	(0.0277)	(0.0123)	(0.0177)
et_3	0.1100***	0.0377	0.0307**	-0.0191
	(0.0230)	(0.0235)	(0.0129)	(0.0192)
Observations	293,442	298,662	313,545	370,506
R^2	0.71726	0.71349	0.71156	0.70131
Within R^2	0.00098	0.00143	0.00175	0.00114
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

7 Appendix: Output Inventory

The streamlined annual writeup now depends on the following output families:

- `annual_filing_core_*`
- `annual_filing_acq_2plus_*`
- `annual_occupancy_2plus_*`
- `annual_pct_*_2plus_*`
- `annual_permit_rate_2plus_*`
- `annual_complaint_rate_2plus_*`
- `annual_rent_core_*`
- `annual_rent_safe_core_*`
- `annual_rental_evidence_1unit_*`

- `annual_rental_evidence_2plus_*`

Legacy annual outputs and scratch stacked CSVs are no longer required for this document.